

Introduction to Contract Theory and Auctions

Reserve Prices

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Reserve Price

A **reserve price** $r > 0$, is a price that the auctioneer puts such that the object is not sold if the winning bid in the auction is less than r . Thus, the seller *reserves the right* to not sell the object if the selling price determined by the auction is less than the threshold amount r . The purpose of such a reserve price is to **discriminate or exclude potential buyers with low valuations**.

- Since the price needs to be at least greater or equal to r , no bidder with value $v < r$ can make a positive profit in the auction.
- Only bidders with values $v \geq r$ will participate.
- In terms of bidding behavior, it is still a **weakly dominant strategy** to bid your own value, even with a reserve price.
- Consider a bidder with value $v = r$, then the expected payment for this bidder is

$$m(r) = G_1^{(n-1)}(r)r$$

- For a bidder with a value $v > r$ the expected payment is now

$$m(v) = G_1^{(n-1)}(r)r + \int_r^v yg_1^{(n-1)}(y)dy$$

- No bidder with value $v < r$ can make a positive profit.
- Consider $\beta : \mathcal{V} \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$, a symmetric equilibrium of the FPA, then we must have that $\beta(r, r) = r$. Why?
- If $v > r$, we have that

$$\begin{aligned}\beta(v, r) &= \mathbb{E}[\max\{Y_1^{(n-1)}, r\} | Y_1^{(n-1)} \leq v] = \frac{1}{G_1^{(n-1)}(v)} \int_0^v \max\{y, r\} g_1^{(n-1)}(y) dy \\ &= \frac{1}{G_1^{(n-1)}(v)} \left[\int_0^r r g_1^{(n-1)}(y) dy + \int_r^v y g_1^{(n-1)}(y) dy \right] \\ &= \frac{1}{G_1^{(n-1)}(v)} \left[r G_1^{(n-1)}(r) + \int_r^v y g_1^{(n-1)}(y) dy \right]\end{aligned}$$

- Thus, when $v \geq r$: $m(v, r) = G_1^{(n-1)}(v) \beta(v, r)$

Interim expected payment with reserve prices

From the previous calculations:

$$m(v, r) = G_1^{(n-1)}(v)\beta(v, r) = rG_1^{(n-1)}(r) + \int_r^v yg_1^{(n-1)}(y)dy \quad \text{if } v \geq r$$

which, not surprisingly, is the same interim expected payment as in the SPA. Now, consider a bidder with value $v = r$, the interim expected payment is:

$$m(r, r) = rG_1^{(n-1)}(r) + \int_r^r yg_1^{(n-1)}(y)dy = rG_1^{(n-1)}(r)$$

Thus, we have that

$$\begin{aligned} m(v, r) &= \begin{cases} rG_1^{(n-1)}(r) + \int_r^v yg_1^{(n-1)}(y)dy & \text{if } v \geq r, \\ 0 & \text{if } v < r. \end{cases} \\ &= \left(rG_1^{(n-1)}(r) + \int_r^v yg_1^{(n-1)}(y)dy \right) I_{\{v \geq r\}}(v) \end{aligned}$$

Revenue Effects of Reserve Prices

The interim expected payment with a reserve price r is:

$m(v, r) = \left(rG_1^{(n-1)}(r) + \int_r^v yg_1^{(n-1)}(y)dy \right) I_{v \geq r}(v)$. Thus:

$$\begin{aligned} E[m(V, r)] &= \int_0^{\bar{v}} \left(rG_1^{(n-1)}(r) + \int_r^v yg_1^{(n-1)}(y)dy \right) I_{v \geq r}(v) f(v) dv \\ &= \int_r^{\bar{v}} \left(rG_1^{(n-1)}(r) + \int_r^v yg_1^{(n-1)}(y)dy \right) f(v) dv \\ &= rG_1^{(n-1)}(r)(1 - F(r)) + \int_r^{\bar{v}} \left(\int_r^v yg_1^{(n-1)}(y)dy \right) f(v) dv \\ &= rG_1^{(n-1)}(r)(1 - F(r)) + \int_r^{\bar{v}} yg_1^{(n-1)}(y) \left(\int_y^{\bar{v}} f(v) dv \right) dy \\ &= rG_1^{(n-1)}(r)(1 - F(r)) + \int_r^{\bar{v}} yg_1^{(n-1)}(y) (1 - F(y)) dy \end{aligned}$$

Optimal Reserve Prices

- Suppose that the seller attaches a value $v_0 \in [0, \bar{v})$ to the object.
- The seller will never use $r < v_0$.
- The expected revenue of a reserve price $r \geq v_0$ is

$$\Pi_0(r) = n \mathbb{E}[m(V, r)] + G_1^{(n)}(r)v_0 = n \mathbb{E}[m(V, r)] + F(r)^n v_0$$

- In order to the optimal reserve price r^* , we need to find $\frac{d\Pi_0}{dr}$:

$$\begin{aligned}\Pi_0 &= nr G_1^{(n-1)}(r) - nr G_1^{(n-1)}(r)F(r) + n \int_r^{\bar{v}} y g_1^{(n-1)}(y) (1 - F(y)) dy + F(r)^n v_0 \\ \frac{d\Pi_0}{dr} &= n \left(G_1^{(n-1)}(r) + r g_1^{(n-1)}(r) \right) - n \left(r G_1^{(n-1)}(r) f(r) + (G_1^{(n-1)}(r) + r g_1^{(n-1)}(r)) F(r) \right) \\ &\quad - nr g_1^{(n-1)}(r) (1 - F(r)) + n F(r)^{n-1} f(r) v_0 \\ &= n G_1^{(n-1)}(r) - nr G_1^{(n-1)}(r) f(r) - n G_1^{(n-1)}(r) F(r) + n F(r)^{n-1} f(r) v_0.\end{aligned}$$

- Multiplying $\frac{d\Pi_0}{dr}$ by $\frac{1-F(r)}{1-F(r)}$ and defining the **hazard rate** as $\lambda(x) = \frac{f(x)}{1-F(x)}$, we obtain that

$$\frac{d\Pi_0}{dr} = nG_1^{(n-1)}(r)(1-F(r))(1-(r-v_0)\lambda(r)).$$

- The first order condition for an interior optimum is then:

$$nG_1^{(n-1)}(r^*)(1-F(r^*))(1-(r^*-v_0)\lambda(r^*)) = 0$$

- The first part of the expression is strictly positive for all r in the interior of $\mathcal{V}$, thus, at an interior optimum we must have:

$$1 = (r^* - v_0)\lambda(r^*) \Rightarrow r^* - \frac{1}{\lambda(r^*)} = v_0 \Rightarrow r^* - \frac{1-F(r^*)}{f(r^*)} = v_0.$$

- If $v_0 = 0$ then

$$r^* = \frac{1-F(r^*)}{f(r^*)}.$$

Two bidders, uniform $[0, 1]$ valuations again

The setup is again that we have two bidders with i.i.d. $U[0, 1]$ valuations, and the auctioneer values the object at $v_0 = 0$. Then in this case

$$r^* = 1 - r^* \Rightarrow r^* = 1/2.$$

Recall that without a reserve price, the expected revenue was $1/3$.

With $r^* = 1/2$ you can check (homework!) that revenues are $5/12$.